

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M. Tech. Electrical Semester – I, Examination, Dec 2013

EE705 Modern Power System Operations

Date: 06/01/2014, Monday

Time: 10.00 AM to 01.00 PM

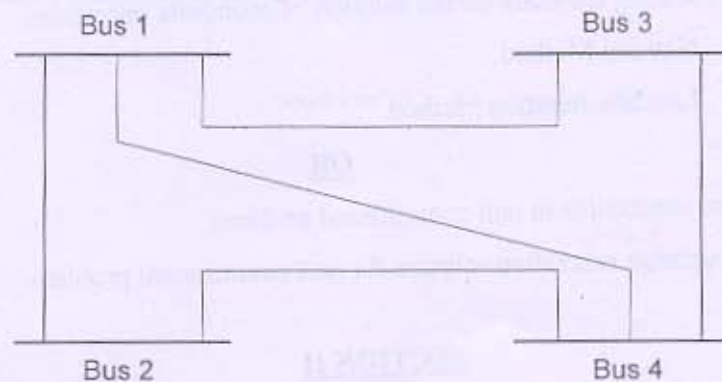
Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each section.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION I

Q - 1 (a)



[07]

Fig. 1

From Bus	To Bus	R (pu)	X (pu)	B/2 (pu)
1	2	0.05	0.15	0.1
1	3	0.04	0.25	0.05
1	4	0.04	0.20	0.15
2	4	0.06	0.15	0.08
3	4	0.05	0.20	0.06

For the system shown in Fig. 1, find bus admittance matrix (Y_{BUS}).

- Q - 2 (a) Draw the flow chart of Gauss-Siedel or Newton-Raphson Method and explain the steps for the solution of load flow considering all kinds of buses. [07]
- (b) Derive the equations for optimal power generations considering the transmission losses and all types of constraints. [07]

OR

Q - 2 (a) State all the assumptions made for FDLF. Draw the flow chart for FDLF and explain the solution steps for load flow analysis. [07]

(b) Input-Output characteristic of three generating units are as follow. [07]

$$F_1 = 0.0006P_{G1}^2 + 0.5P_{G1} + 6 \text{ Btu/h}$$

$$F_2 = 0.0005P_{G2}^2 + 0.6P_{G2} + 5 \text{ Btu/h}$$

$$F_3 = 0.0007P_{G3}^2 + 0.4P_{G3} + 3 \text{ Btu/h}$$

Determine economic operation point for three generating units when delivering a total of 500 MW. Find the solution with the use of gradient method.

Q - 3 (a) What is optimal power flow? Explain all the steps of Gradient Method in detail for the solution of OPF considering all buses in a system. [07]

(b) Explain following methods for the solution of economic operations. [07]

(i) Newton Method

(ii) Lambda-Iteration Method

OR

(a) Discuss the constraints in unit commitment problem [07]

(b) Explain Lagrange relaxation solution for unit commitment problem [07]

SECTION II

Q - 4 (a) Explain clearly the application and need of state estimation in power system with a diagram. [07]

Q - 5 (a) Explain maximum likelihood weighted least square method for state estimation considering two measurements. [07]

(b) Discuss the state estimation of an AC network with flowchart. [07]

OR

Q - 5 (a) Derive line outage distribution factor for power system security [07]

(b) Discuss the following in relation with power system security evaluation [07]

(i) System Monitoring

(ii) Contingency Analysis

(iii) Security Constrained Optimal Power Flows

Q - 6 (a) Derive the mathematical model of following components for AGC [08]

(i) Generator Model

- (ii) Tie-line Model. Derive the expression for change in tie line power for step change in load demand
- (b) Explain the Automatic Generation Control w.r.t. following [06]
- (a) Supplementary Control Action
 - (b) Tie-line Control

OR

- (a) Write a short note on demand side management. [07]
- (b) Two generators rated 250 MW and 400 MW are operating in parallel. The droop characteristics of the governors are 4% and 6%, respectively. How would a load of 650 MW be shared between them? What will be the system frequency? Assume nominal system frequency is 60 Hz and no governing action. [07]
